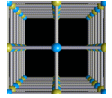


Finite element approximation of nonlocal fracture models

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Joint work with
Dr. Robert Lipton

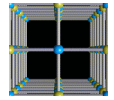
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Outline of talk

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- Peridynamic: Introduction
- Well-posedness of Peridynamic solutions
- A priori error estimates on finite element approximations
- Numerical verification
- Future works



Introduction

2

Let D be the material domain, D_c be nonlocal boundary, and $\mathbf{u}$ be the displacement field.

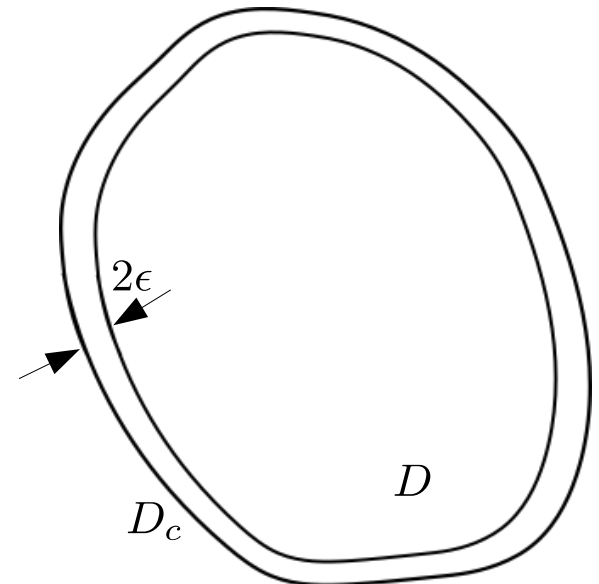
Let $\mathbf{x}$ denote the material point and $\chi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$ is the deformed position.

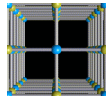
Strain between two material point $\mathbf{x}$ and $\mathbf{y}$ is given by

$$S(\mathbf{y}, \mathbf{x}; \mathbf{u}) = \frac{|\mathbf{y} + \mathbf{u}(\mathbf{y}) - \mathbf{x} - \mathbf{u}(\mathbf{x})| - |\mathbf{y} - \mathbf{x}|}{|\mathbf{y} - \mathbf{x}|}$$

Assuming that displacement is small compared to the size of material, we linearize S and get

$$S(\mathbf{y}, \mathbf{x}; \mathbf{u}) = \frac{\mathbf{u}(\mathbf{y}) - \mathbf{u}(\mathbf{x})}{|\mathbf{y} - \mathbf{x}|} \cdot \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|}$$





Introduction: Generic force

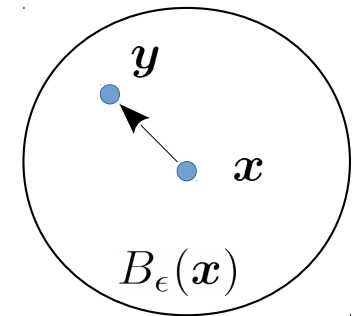
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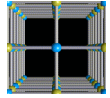
Consider a material point $\mathbf{x}$. We introduce a length scale ϵ which is called size of horizon. This controls the extent of nonlocal interaction in the material.

Generic form of force at $\mathbf{x}$ in peridynamic model is given by

$$\mathbf{f}^\epsilon(\mathbf{x}; \mathbf{u}) = \frac{1}{|B_\epsilon(\mathbf{x})|} \int_{B_\epsilon(\mathbf{x})} \hat{\mathbf{f}}^\epsilon(\mathbf{y}, \mathbf{x}; \mathbf{u}) d\mathbf{y}$$

$\hat{\mathbf{f}}^\epsilon$ depends on choice of ϵ .





Introduction: Example of a bond-force

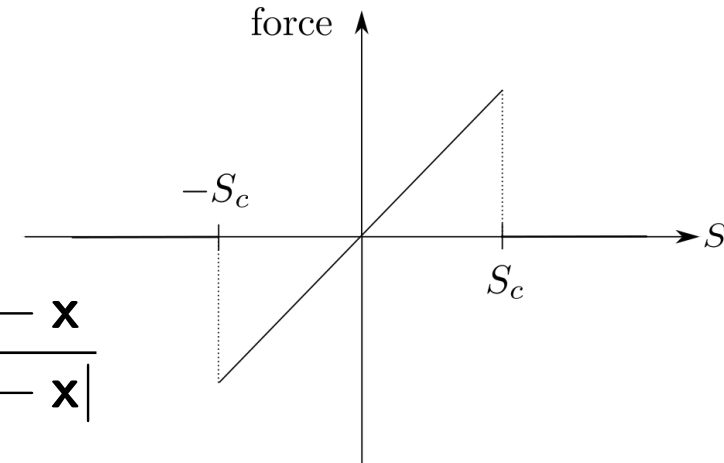
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Example:

$$\hat{\mathbf{f}}^\epsilon(\mathbf{y}, \mathbf{x}, \mathbf{u}) = \mu(S(\mathbf{y}, \mathbf{x}; \mathbf{u})) 4 \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon} S(\mathbf{y}, \mathbf{x}; \mathbf{u}) \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|}$$

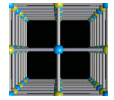
where $\mu(S) = 1$ if $|S| < S_c$ and $\mu(S) = 0$ when $|S| \geq S_c$.



If $\mathbf{u} \in C^3(D; \mathbb{R}^d)$, and $\sup_{\mathbf{x} \in D} |\nabla^3 \mathbf{u}(\mathbf{x})| < \infty$ then

$$\sup_{x \in D} |\mathbf{f}^\epsilon(\mathbf{x}; \mathbf{u}) - \nabla \cdot \bar{\mathbb{C}} \mathcal{E} \mathbf{u}(\mathbf{x})| = O(\epsilon^2), \quad \bar{\mathbb{C}} = \frac{2}{|B_1(\mathbf{0})|} \int_{B_1(\mathbf{0})} J(|\boldsymbol{\xi}|) \mathbf{e}_{\boldsymbol{\xi}} \otimes \mathbf{e}_{\boldsymbol{\xi}} \otimes \mathbf{e}_{\boldsymbol{\xi}} \otimes \mathbf{e}_{\boldsymbol{\xi}} |\boldsymbol{\xi}| d\boldsymbol{\xi},$$

$\mathbf{e}_{\boldsymbol{\xi}} = \boldsymbol{\xi}/|\boldsymbol{\xi}|$ and the strain tensor is $\mathcal{E} \mathbf{u}(\mathbf{x}) = (\nabla \mathbf{u}(\mathbf{x}) + \nabla \mathbf{u}^T(\mathbf{x}))/2$.



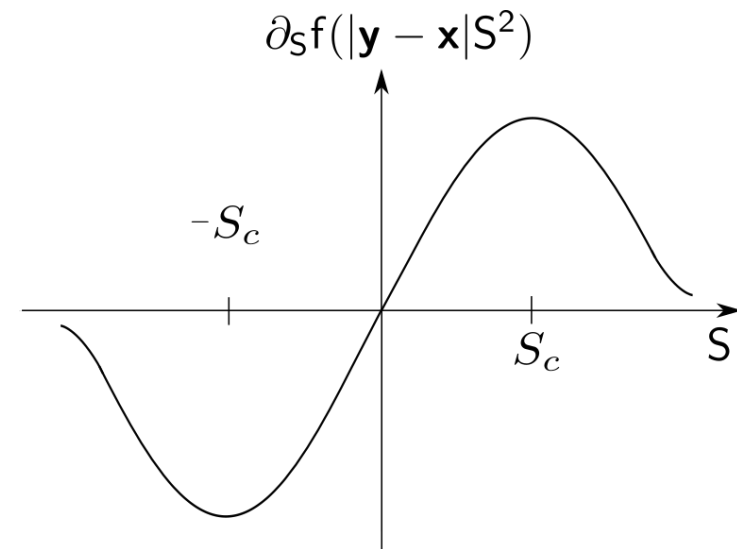
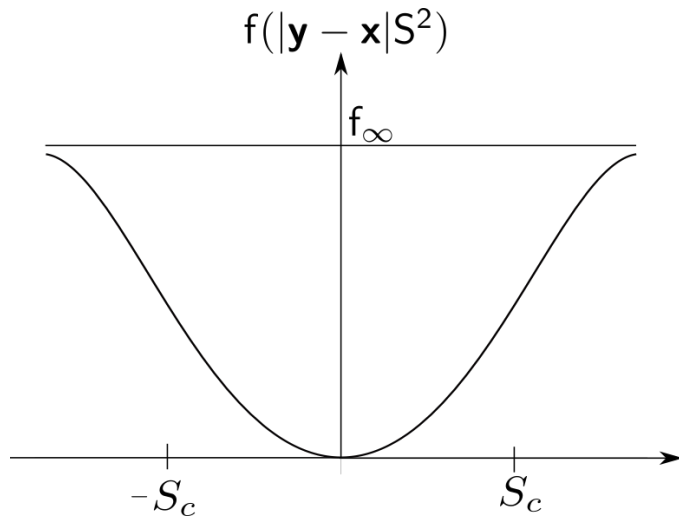
Introduction: Regularized force

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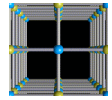
- We consider peridynamic force of the form

$$\mathbf{f}^\epsilon(\mathbf{x}; \mathbf{u}) = \frac{2}{|B_\epsilon(\mathbf{0})|} \int_{B_\epsilon(\mathbf{x})} \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon |\mathbf{y} - \mathbf{x}|} \partial_S f(|\mathbf{y} - \mathbf{x}| S^2) \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|} d\mathbf{y}$$

where f is smooth, bounded far away, and linear near origin (**Lipton 2014**¹)



- Critical strain: $S_c(\mathbf{y}, \mathbf{x}) = \frac{\bar{r}}{\sqrt{|\mathbf{y} - \mathbf{x}|}}$

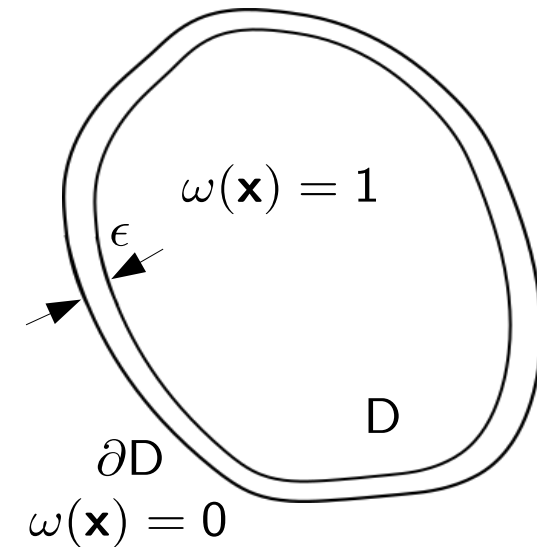


Introduction: Regularized force

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- In **Jha & Lipton 2017a**¹ and **Jha & Lipton 2017b**², we introduce boundary function ω in peridynamic force as follows

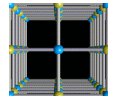
$$\mathbf{f}^\epsilon(\mathbf{x}; \mathbf{u}) = \frac{2}{|B_\epsilon(\mathbf{0})|} \int_{B_\epsilon(\mathbf{x})} \omega(\mathbf{x})\omega(\mathbf{y}) \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon|\mathbf{y} - \mathbf{x}|} \partial_S f(|\mathbf{y} - \mathbf{x}|S^2) \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|} d\mathbf{y}$$



- With boundary function ω , we can show existence of solutions in regular spaces like $C_0^{0,\gamma}(D; \mathbb{R}^d)$ and $H_0^2(D; \mathbb{R}^d) \cup L^\infty(D; \mathbb{R}^d)$ for Dirichlet boundary condition $\mathbf{u} = \mathbf{0}$ on ∂D .

[1] Prashant K. Jha and Robert Lipton (2017) Numerical analysis of peridynamic models in Hölder space. To appear in SIAM Journal on Numerical Analysis. arXiv preprint arXiv:1701.02818

[2] Prashant K. Jha and Robert Lipton (2017) Finite element approximation of nonlocal fracture models. Under review in IMA Journal of Numerical Analysis. arXiv preprint arXiv:1710.07661



Introduction: Equation of motion

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- Peridynamics equation: for $\mathbf{x} \in D$ and $t \in [0, T]$

$$\rho \ddot{\mathbf{u}}(\mathbf{x}, t) = \mathbf{f}^\epsilon(\mathbf{x}; \mathbf{u}(t)) + \mathbf{b}(\mathbf{x}, t)$$

- Boundary condition: $\mathbf{u}(\mathbf{x}, t) = \mathbf{0}$ for $\mathbf{x} \in \partial D$ and for $t \in [0, T]$

- Initial condition: $\mathbf{u}(\mathbf{x}, 0) = \mathbf{u}_0(\mathbf{x})$ and $\dot{\mathbf{u}}(\mathbf{x}, 0) = \mathbf{v}_0(\mathbf{x})$ for $\mathbf{x} \in D$

- Weak form: Multiplying peridynamic equation by smooth test function $\tilde{\mathbf{u}}$ such that $\tilde{\mathbf{u}} = \mathbf{0}$ on ∂D , integrating over D , and using nonlocal integration by parts, gives

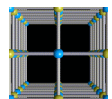
$$(\rho \ddot{\mathbf{u}}(t), \tilde{\mathbf{u}}) + a^\epsilon(\mathbf{u}(t), \tilde{\mathbf{u}}) = (\mathbf{b}(t), \tilde{\mathbf{u}})$$

where

$$a^\epsilon(\mathbf{u}, \mathbf{v}) = \frac{2}{\epsilon |B_\epsilon(\mathbf{x})|} \int_D \int_{B_\epsilon(\mathbf{x})} \omega(\mathbf{x}) \omega(\mathbf{y}) J^\epsilon(|\mathbf{y} - \mathbf{x}|) f'(|\mathbf{y} - \mathbf{x}| S(\mathbf{u})^2) |\mathbf{y} - \mathbf{x}| S(\mathbf{u}) S(\mathbf{v}) d\mathbf{y} d\mathbf{x}$$

and

$$S(\mathbf{u}) = \frac{\mathbf{u}(\mathbf{y}) - \mathbf{u}(\mathbf{x})}{|\mathbf{y} - \mathbf{x}|} \cdot \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|}, S(\mathbf{v}) = \frac{\mathbf{v}(\mathbf{y}) - \mathbf{v}(\mathbf{x})}{|\mathbf{y} - \mathbf{x}|} \cdot \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|}$$

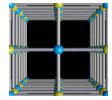


Finite element approximation

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We approximate peridynamic equation using linear continuous finite elements. We focus on following three key points

- Well-posedness of peridynamic equation in $H_0^2(D; \mathbb{R}^d)$ space.
- A priori error estimates due to finite element approximations for exact solutions in $H_0^2(D; \mathbb{R}^d)$.
- Numerical verifications of convergence rate.



Well-posedness of peridynamic equation

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Let W denote the $H_0^2(D; \mathbb{R}^d) \cap L^\infty(D; \mathbb{R}^d)$ space. Norm on W is defined as

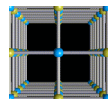
$$\|\mathbf{u}\|_W := \|\mathbf{u}\|_2 + \|\mathbf{u}\|_\infty$$

We will assume that $\mathbf{u} \in H_0^2(D; \mathbb{R}^d)$ is extended by zero outside D , therefore, $\mathbf{u} = \mathbf{0}$, $\nabla \mathbf{u} = \mathbf{0}$, $\nabla^2 \mathbf{u} = \mathbf{0}$ for $\mathbf{x} \notin D$ and $\|\mathbf{u}\|_{H^2(D; \mathbb{R}^d)} = \|\mathbf{u}\|_{H^2(\mathbb{R}^d; \mathbb{R}^d)}$.

To show existence of solutions in W , we proceed as follows:

- Obtain Lipschitz bound on peridynamic force in W .
- Using Lipschitz bound, show local existence of unique solutions.

Show that local existence of unique solutions can be repeatedly applied to get global existence of solutions for any time domain $(-T, T)$.



Well-posedness of peridynamic equation

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We write the peridynamics equation as an equivalent first order system with $y_1(t) = \mathbf{u}(t)$ and $y_2(t) = \mathbf{v}(t)$ with $\mathbf{v}(t) = \dot{\mathbf{u}}(t)$. Let $y = (y_1, y_2)^T$ where $y_1, y_2 \in W$ and let $F^\epsilon(y, t) = (F_1^\epsilon(y, t), F_2^\epsilon(y, t))^T$ such that

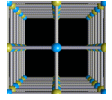
$$F_1^\epsilon(y, t) := y_2,$$

$$F_2^\epsilon(y, t) := \mathbf{f}^\epsilon(y_1) + \mathbf{b}(t).$$

The initial boundary value is equivalent to the initial boundary value problem for the first order system given by

$$\dot{y}(t) = F^\epsilon(y, t),$$

with initial condition given by $y(0) = (\mathbf{u}_0, \mathbf{v}_0)^T \in W \times W$.



Lipschitz bound on peridynamic force

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Theorem 1. *Lipschitz bound on peridynamics force*

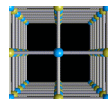
Assuming $|\nabla\omega| \leq C_{\omega_1} < \infty$, $|\nabla^2\omega| \leq C_{\omega_2} < \infty$, for any $\mathbf{u}, \mathbf{v} \in W$, we have

$$\|\mathbf{f}^\epsilon(\mathbf{u}) - \mathbf{f}^\epsilon(\mathbf{v})\|_W \leq \frac{\bar{L}_1 + \bar{L}_2(\|\mathbf{u}\|_W + \|\mathbf{v}\|_W) + \bar{L}_3(\|\mathbf{u}\|_W + \|\mathbf{v}\|_W)^2}{\epsilon^3} \|\mathbf{u} - \mathbf{v}\|_W$$

where constants $\bar{L}_1, \bar{L}_2, \bar{L}_3$ are independent of ϵ , $\mathbf{u}$, and $\mathbf{v}$. Also, for $\mathbf{u} \in W$, we have

$$\|\mathbf{f}^\epsilon(\mathbf{u})\|_W \leq \frac{\bar{L}_4\|\mathbf{u}\|_W + \bar{L}_5\|\mathbf{u}\|_W^2}{\epsilon^{5/2}},$$

where constants are independent of ϵ and $\mathbf{u}$.



Local existence

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Theorem 2. *Local existence and uniqueness*

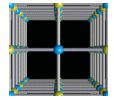
Given $X = W \times W$, $\mathbf{b}(t) \in W$, and initial data $x_0 = (\mathbf{u}_0, \mathbf{v}_0) \in X$. We suppose that $\mathbf{b}(t)$ is continuous in time over some time interval $I_0 = (-T, T)$ and satisfies $\sup_{t \in I_0} \|\mathbf{b}(t)\|_W < \infty$. Then, there exists a time interval $I' = (-T', T') \subset I_0$ and unique solution $y = (y^1, y^2)$ such that $y \in C^1(I'; X)$ and

$$y(t) = x_0 + \int_0^t F^\epsilon(y(\tau), \tau) d\tau, \text{ for } t \in I'$$

or equivalently

$$y'(t) = F^\epsilon(y(t), t), \text{ with } y(0) = x_0, \text{ for } t \in I'$$

where $y(t)$ and $y'(t)$ are Lipschitz continuous in time for $t \in I' \subset I_0$.



Local existence

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Proof: Let $T' > 0$ and $Y(T')$ be a set of functions $y(t) \in W$ for $t \in (-T', T')$. We show that there exists such a set $Y(T')$ and $T' > 0$ such that map S_{x_0} , defined as follows

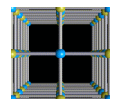
$$S_{x_0}(y)(t) = x_0 + \int_0^t F^\epsilon(y(\tau), \tau) d\tau,$$

or in element form

$$S_{x_0}^1(y)(t) = x_0^1 + \int_0^t y^2(\tau) d\tau$$

$$S_{x_0}^2(y)(t) = x_0^2 + \int_0^t (\mathbf{f}^\epsilon(y^1(\tau)) + \mathbf{b}(\tau)) d\tau,$$

maps functions in $Y(T')$ to functions in $Y(T')$. We then apply fixed point theorem such as in **Driver 2003**¹.



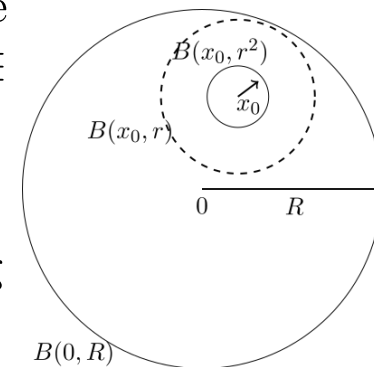
Local existence

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Write $y(t) = (y^1(t), y^2(t))^T$ with $\|y\|_X = \|y^1(t)\|_W + \|y^2(t)\|_W$. Let $R > \|x_0\|_X$ and $B(0, R) = \{y \in X : \|y\|_X < R\}$. Let $r < \min\{1, R - \|x_0\|_X\}$. We have $r^2 < (R - \|x_0\|_X)^2$ and $r^2 < r < R - \|x_0\|_X$. Consider the ball $B(x_0, r^2) = \{y \in X : \|y - x_0\|_X < r^2\}$.

Then we have $B(x_0, r^2) \subset B(x_0, r) \subset B(0, R)$.

Introduce $0 < T' < T$ and the associated set $Y(T')$ of functions in W taking values in $B(x_0, r^2)$ for $I' = (-T', T') \subset I_0 = (-T, T)$. I.e. for all $y \in Y(T')$, $y(t) \in B(x_0, r^2)$ for all $t \in (-T', T')$. We want to find T' such that $S_{x_0}(y)(t) \in B(x_0, r^2)$ for all $t \in (-T', T')$ implying $S_{x_0}(y) \in Y(T')$.



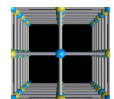
Writing out the transformation with $y(t) \in Y(T')$ gives

$$S_{x_0}^1(y)(t) = x_0^1 + \int_0^t y^2(\tau) d\tau$$

$$S_{x_0}^2(y)(t) = x_0^2 + \int_0^t (f^\epsilon(y^1(\tau)) + b(\tau)) d\tau.$$

We simply have

$$\|S_{x_0}^1(y)(t) - x_0^1\|_W \leq \sup_{t \in (-T', T')} \|y^2(t)\|_W T'.$$



Local existence

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Using bound on $\mathbf{f}^\epsilon$, we have

$$\|S_{x_0}^2(y)(t) - x_0^2\|_W \leq \int_0^t \left[\frac{\bar{L}_4}{\epsilon^{5/2}} \|y^1(\tau)\|_W + \frac{\bar{L}_5}{\epsilon^{5/2}} \|y^1(\tau)\|_W^2 + \|\mathbf{b}(\tau)\|_W \right] d\tau.$$

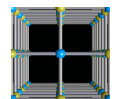
Let $\bar{b} = \sup_{t \in I_0} \|\mathbf{b}(t)\|_W$. Noting that transformation S_{x_0} is defined for $t \in I' = (-T', T')$ and $y(\tau) = (y^1(\tau), y^2(\tau)) \in B(x_0, r^2) \subset B(0, R)$ as $y \in Y(T')$, we have from 3 and 4

$$\|S_{x_0}^1(y)(t) - x_0^1\|_W \leq RT',$$

$$\|S_{x_0}^2(y)(t) - x_0^2\|_W \leq \left[\frac{\bar{L}_4 R + \bar{L}_5 R^2}{\epsilon^{5/2}} + \bar{b} \right] T'.$$

Combining to get

$$\|S_{x_0}(y)(t) - x_0\|_X \leq \left[\frac{\bar{L}_4 R + \bar{L}_5 R^2}{\epsilon^{5/2}} + R + \bar{b} \right] T'.$$



Local existence

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Choosing T' as follow: $T' < \frac{r^2}{\left[\frac{\bar{L}_4 R + \bar{L}_5 R^2}{\epsilon^{5/2}} + R + \bar{b} \right]}$

Then $S_{x_0}(y) \in Y(T')$ for all $y \in Y(T')$ as $\|S_{x_0}(y)(t) - x_0\|_X < r^2$.

Since $r^2 < (R - \|x_0\|_X)^2$, we have

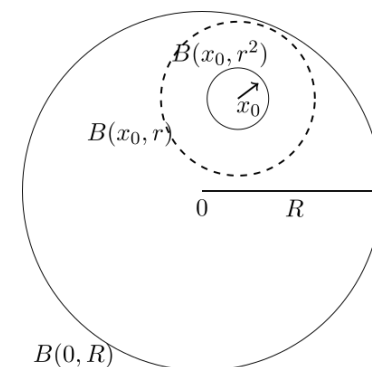
$$T' < \frac{r^2}{\left[\frac{\bar{L}_4 R + \bar{L}_5 R^2}{\epsilon^{5/2}} + R + \bar{b} \right]} < \frac{(R - \|x_0\|_X)^2}{\left[\frac{\bar{L}_4 R + \bar{L}_5 R^2}{\epsilon^{5/2}} + R + \bar{b} \right]}.$$

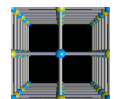
Let $\theta(R)$ be given by

$$\theta(R) := \frac{(R - \|x_0\|_X)^2}{\left[\frac{\bar{L}_4 R + \bar{L}_5 R^2}{\epsilon^{5/2}} + R + \bar{b} \right]}.$$

Note that $\theta(R)$ is increasing with $R > 0$ and satisfies

$$\theta_\infty := \lim_{R \rightarrow \infty} \theta(R) = \frac{\epsilon^{5/2}}{\bar{L}_5}.$$





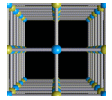
Local existence

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So given R and $\|x_0\|_X$ we choose T' according to

$$\frac{\theta(R)}{2} < T' < \theta(R),$$

and set $I' = (-T', T')$. This way we have shown that for time domain I' the transformation $S_{x_0}(y)(t)$ maps $Y(T')$ into itself. Existence and uniqueness of solution can be established using Theorem 6.10 in **Driver 2003**¹.



Global existence

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Theorem 3. *Existence and uniqueness of solutions over finite time intervals*

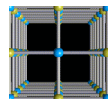
For any initial condition $x_0 \in X = W \times W$, time interval $I_0 = (-T, T)$, and right hand side $\mathbf{b}(t)$ continuous in time for $t \in I_0$ such that $\mathbf{b}(t)$ satisfies $\sup_{t \in I_0} \|\mathbf{b}(t)\|_W < \infty$, there is a unique solution $y(t) \in C^1(I_0; X)$ of

$$y(t) = x_0 + \int_0^t F^\epsilon(y(\tau), \tau) d\tau,$$

or equivalently

$$y'(t) = F^\epsilon(y(t), t), \text{ with } y(0) = x_0,$$

where $y(t)$ and $y'(t)$ are Lipschitz continuous in time for $t \in I_0$.



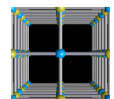
Global existence

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We have shown a unique local solution over a time domain $(-T', T')$ with $\frac{\theta(R)}{2} < T'$. Since $\theta(R) \nearrow \epsilon^{5/2}/\bar{L}_5$ as $R \nearrow \infty$ we can fix a tolerance $\eta > 0$ so that $[(\epsilon^{5/2}/2\bar{L}_5) - \eta] > 0$.

Then for any initial condition in W and $b = \sup_{t \in [-T, T)} \|\mathbf{b}(t)\|_W$ we can choose R sufficiently large so that $\|x_0\|_X < R$ and $0 < (\epsilon^{5/2}/2\bar{L}_5) - \eta < T'$.

Since choice of T' is independent of initial condition and R , we can always find local solutions for time intervals $(-T', T')$ for T' larger than $[(\epsilon^{5/2}/2\bar{L}_5) - \eta] > 0$. Therefore we apply the local existence and uniqueness result to uniquely continue local solutions up to an arbitrary time interval $(-T, T)$.



Finite element approximation

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Let V_h be the approximation of $H_0^2(D, \mathbb{R}^d)$ associated to the linear continuous interpolation function over triangulation $\mathcal{T}_h$ where h denotes the size of finite element mesh. Let $\mathcal{I}_h(\mathbf{u})$ be defined as below

$$\mathcal{I}_h(\mathbf{u})(\mathbf{x}) = \sum_{T \in \mathcal{T}_h} \left[\sum_{i \in N_T} \mathbf{u}(\mathbf{x}_i) \phi_i(\mathbf{x}) \right].$$

Assuming that the size of each element in triangulation $\mathcal{T}_h$ is bounded by h , we have (see Theorem 4.6 **Arnold 2011**¹)

$$\|\mathbf{u} - \mathcal{I}_h(\mathbf{u})\| \leq ch^2 \|\mathbf{u}\|_2, \quad \forall \mathbf{u} \in H_0^2(D; \mathbb{R}^d).$$

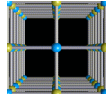
Projection of function in FE space:

$$\|\mathbf{u} - \mathbf{r}_h(\mathbf{u})\| = \inf_{\tilde{\mathbf{u}} \in V_h} \|\mathbf{u} - \tilde{\mathbf{u}}\|.$$

We have

$$(\mathbf{r}_h(\mathbf{u}), \tilde{\mathbf{u}}) = (\mathbf{u}, \tilde{\mathbf{u}}), \quad \forall \tilde{\mathbf{u}} \in V_h.$$

$$\|\mathbf{u} - \mathbf{r}_h(\mathbf{u})\| \leq ch^2 \|\mathbf{u}\|_2 \quad \forall \mathbf{u} \in H_0^2(D; \mathbb{R}^d).$$



Semi-discrete approximation and stability

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Let $\mathbf{u}_h(t) \in V_h$ be the approximation of $\mathbf{u}(t)$ which satisfies following

$$(\ddot{\mathbf{u}}_h, \tilde{\mathbf{u}}) + a^\epsilon(\mathbf{u}_h(t), \tilde{\mathbf{u}}) = (\mathbf{b}(t), \tilde{\mathbf{u}}), \quad \forall \tilde{\mathbf{u}} \in V_h.$$

We show that the semi-discrete approximation is stable, i.e. energy at time t is bounded by initial energy and work done by the body force.

The total energy $\mathcal{E}^\epsilon(\mathbf{u})(t)$ is given by the sum of kinetic and potential energy given by

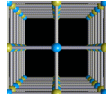
$$\mathcal{E}^\epsilon(\mathbf{u})(t) = \frac{1}{2} \|\dot{\mathbf{u}}(t)\|_{L^2}^2 + PD^\epsilon(\mathbf{u}(t)), PD^\epsilon(\mathbf{u}) = \int_D \left[\frac{1}{|B_\epsilon(\mathbf{x})|} \int_{B_\epsilon(\mathbf{x})} W^\epsilon(S(\mathbf{u}), \mathbf{y} - \mathbf{x}) d\mathbf{y} \right] d\mathbf{x},$$

where bond-potential is given by $W^\epsilon(S, \mathbf{y} - \mathbf{x}) = \omega(\mathbf{x})\omega(\mathbf{y}) \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon} f(|\mathbf{y} - \mathbf{x}|S^2)$.

Theorem 4. Stability of semi-discrete approximation

The semi-discrete scheme is stable and the energy $\mathcal{E}^\epsilon(\mathbf{u}_h)(t)$ satisfies the following bound

$$\mathcal{E}^\epsilon(\mathbf{u}_h)(t) \leq \left[\sqrt{\mathcal{E}^\epsilon(\mathbf{u}_h)(0)} + \int_0^t \|\mathbf{b}(\tau)\| d\tau \right]^2.$$



Central difference time discretization

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$(\mathbf{u}_h^k, \mathbf{v}_h^k)$ and $(\mathbf{u}^k, \mathbf{v}^k)$ denote the approximate and the exact solution at k^{th} step. Projection is denoted as $(\mathbf{r}_h(\mathbf{u}^k), \mathbf{r}_h(\mathbf{v}^k))$. Approximate initial condition $\mathbf{u}_0, \mathbf{v}_0$ by their projection $\mathbf{r}_h(\mathbf{u}_0), \mathbf{r}_h(\mathbf{v}_0)$ and set $\mathbf{u}_h^0 = \mathbf{r}_h(\mathbf{u}_0), \mathbf{v}_h^0 = \mathbf{r}_h(\mathbf{v}_0)$.

For $k \geq 1$, $(\mathbf{u}_h^k, \mathbf{v}_h^k)$ satisfies, for all $\tilde{\mathbf{u}} \in V_h$,

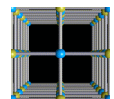
$$\begin{aligned} \left(\frac{\mathbf{u}_h^{k+1} - \mathbf{u}_h^k}{\Delta t}, \tilde{\mathbf{u}} \right) &= (\mathbf{v}_h^{k+1}, \tilde{\mathbf{u}}), \\ \left(\frac{\mathbf{v}_h^{k+1} - \mathbf{v}_h^k}{\Delta t}, \tilde{\mathbf{u}} \right) &= (\mathbf{f}^\epsilon(\mathbf{u}_h^k), \tilde{\mathbf{u}}) + (\mathbf{b}_h^k, \tilde{\mathbf{u}}), \end{aligned}$$

where we denote projection of $\mathbf{b}(\mathbf{t}^k)$, $\mathbf{r}_h(\mathbf{b}(\mathbf{t}^k))$, as $\mathbf{b}_h^k$. Combining the two equations delivers central difference equation for $\mathbf{u}_h^k$. We have

$$\left(\frac{\mathbf{u}_h^{k+1} - 2\mathbf{u}_h^k + \mathbf{u}_h^{k-1}}{\Delta t^2}, \tilde{\mathbf{u}} \right) = (\mathbf{f}^\epsilon(\mathbf{u}_h^k), \tilde{\mathbf{u}}) + (\mathbf{b}_h^k, \tilde{\mathbf{u}}), \quad \forall \tilde{\mathbf{u}} \in V_h.$$

For $k = 0$, we have $\forall \tilde{\mathbf{u}} \in V_h$

$$\left(\frac{\mathbf{u}_h^1 - \mathbf{u}_h^0}{\Delta t^2}, \tilde{\mathbf{u}} \right) = \frac{1}{2}(\mathbf{f}^\epsilon(\mathbf{u}_h^0), \tilde{\mathbf{u}}) + \frac{1}{\Delta t}(\mathbf{v}_h^0, \tilde{\mathbf{u}}) + \frac{1}{2}(\mathbf{b}_h^0, \tilde{\mathbf{u}}).$$



Convergence of approximation

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Error E^k is given by $E^k := \|\mathbf{u}_h^k - \mathbf{u}(t^k)\| + \|\mathbf{v}_h^k - \mathbf{v}(t^k)\|$. We split the error as follows

$$E^k \leq (\|\mathbf{u}^k - \mathbf{r}_h(\mathbf{u}^k)\| + \|\mathbf{v}^k - \mathbf{r}_h(\mathbf{v}^k)\|) + (\|\mathbf{r}_h(\mathbf{u}^k) - \mathbf{u}_h^k\| + \|\mathbf{r}_h(\mathbf{v}^k) - \mathbf{v}_h^k\|),$$

where first term is error between exact solution and projections, and second term is error between projections and approximate solution.

Let

$$\mathbf{e}_h^k(u) := \mathbf{r}_h(\mathbf{u}^k) - \mathbf{u}_h^k, \mathbf{e}_h^k(v) := \mathbf{r}_h(\mathbf{v}^k) - \mathbf{v}_h^k$$

and

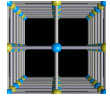
$$e^k := \|\mathbf{e}_h^k(u)\| + \|\mathbf{e}_h^k(v)\|.$$

We have

$$E^k \leq C_p h^2 + e^k,$$

where

$$C_p := c \left[\sup_t \|\mathbf{u}(t)\|_2 + \sup_t \left\| \frac{\partial \mathbf{u}(t)}{\partial t} \right\|_2 \right].$$



Convergence of approximation

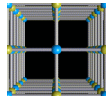
24

Theorem 5. *Convergence of Central difference approximation*

Let $(\mathbf{u}, \mathbf{v})$ be the exact solution of peridynamics equation and Let $(\mathbf{u}_h^k, \mathbf{v}_h^k)$ be the FE approximate solution. If $\mathbf{u}, \mathbf{v} \in C^2([0, T], H_0^2(D; \mathbb{R}^d))$, then the scheme is consistent and the error E^k satisfies following bound

$$\sup_{k \leq T/\Delta t} E^k = C_t \Delta t + C_s \frac{h^2}{\epsilon^2}$$

where constant C_t and C_s are independent of h and Δt and depends on the norm of exact solution. Constant L/ϵ^2 is the Lipschitz constant of $\mathbf{f}^\epsilon(\mathbf{u})$ in L^2 .



Convergence of approximation

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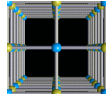
Outline of proof:

(1) Write peridynamics equation for projection $(\mathbf{r}_h(\mathbf{u}^k), \mathbf{r}_h(\mathbf{v}^k))$ which involves consistency error.

(2) Estimate consistency error terms. One of the error term is as follows

$$\|\mathbf{f}^\epsilon(\mathbf{u}^k) - \mathbf{f}^\epsilon(\mathbf{r}_h(\mathbf{u}^k))\| \leq \frac{L}{\epsilon^2} \|\mathbf{u}^k - \mathbf{r}_h(\mathbf{u}^k)\|_{L^2} \leq \frac{Lc}{\epsilon^2} h^2 \sup_t \|\mathbf{u}(t)\|_2$$

(3) Subtract peridynamics equation corresponding to projection $(\mathbf{r}_h(\mathbf{u}^k), \mathbf{r}_h(\mathbf{v}^k))$ and approximate solution $(\mathbf{u}_h^k, \mathbf{v}_h^k)$, use estimates on consistency errors, and apply discrete Grönwall inequality to obtain the bound on $\mathbf{e}^k = \|\mathbf{u}_h^k - \mathbf{r}_h(\mathbf{u}^k)\| + \|\mathbf{v}_h^k - \mathbf{r}_h(\mathbf{v}^k)\|$.



Stability of fully discrete approximation: Linearized peridynamic equation

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Consider linearization of peridynamic force $\mathbf{f}^\epsilon$ defined as

$$\mathbf{f}_l^\epsilon(\mathbf{u})(\mathbf{x}) = \frac{4}{|B_\epsilon(\mathbf{x})|} \int_{B_\epsilon(\mathbf{x})} \omega(\mathbf{x})\omega(\mathbf{y}) \frac{J^\epsilon(|\mathbf{y} - \mathbf{x}|)}{\epsilon} f'(0) S(\mathbf{u}) \frac{\mathbf{y} - \mathbf{x}}{|\mathbf{y} - \mathbf{x}|} d\mathbf{y}.$$

Weak form of peridynamic equation is given by $(\rho \ddot{\mathbf{u}}(t), \tilde{\mathbf{u}}) + \mathbf{a}_l^\epsilon(\mathbf{u}(t), \tilde{\mathbf{u}}) = (\mathbf{b}(t), \tilde{\mathbf{u}})$, where $\mathbf{a}_l^\epsilon(\mathbf{u}, \mathbf{v})$ is now bilinear map.

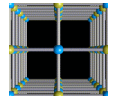
Following **Karaa 2012**¹, we have

Theorem 6. *Stability of Central difference approximation of linearized peridynamics*

In the absence of body force $\mathbf{b}(t) = \mathbf{0}$ for all t , if Δt satisfies the CFL like condition

$$\frac{\Delta t^2}{4} \sup_{\mathbf{u} \in V_h \setminus \{0\}} \frac{\mathbf{a}_l^\epsilon(\mathbf{u}, \mathbf{u})}{(\mathbf{u}, \mathbf{u})} \leq 1,$$

then the discrete energy is conserved and we have the stability.



Numerical results: Exact solutions¹

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- $\epsilon = 0.2$
- $h = \epsilon/4, \epsilon/8$
- Time domain $[0, 1]$ with $\Delta t = 2 \times 10^{-5}$
- $\rho = 1, f(r) = 1 - \exp[-r], J(r) = 1 - r$

Let $\mathbf{w}(\mathbf{x}, t) = a(\mathbf{x}) \sin(n\pi(\mathbf{d} \cdot \mathbf{x} + t))\mathbf{d}$,

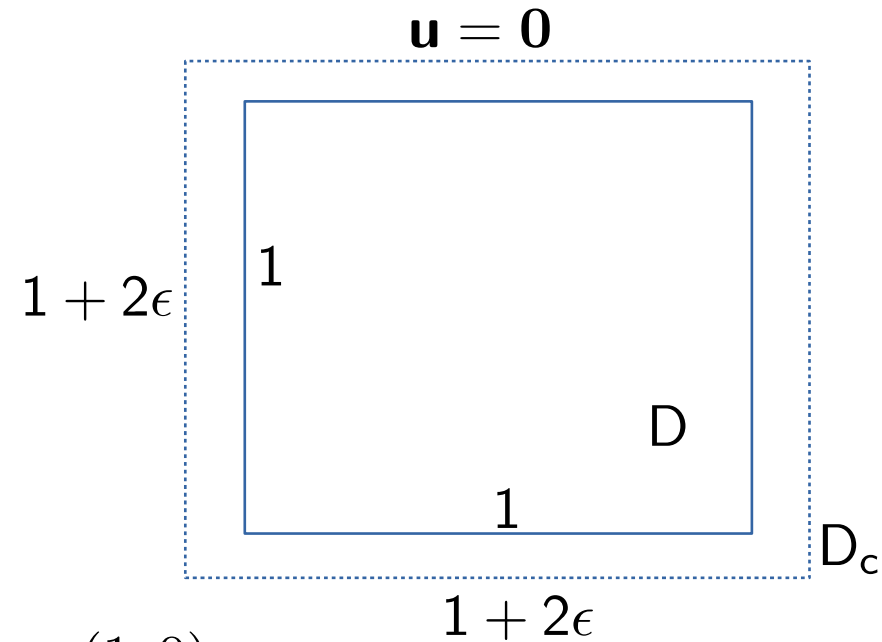
where $a(\mathbf{x}) = 0.001 * x_1 x_2 (1 - x_1)(1 - x_2), \mathbf{d} = (1, 0)$.

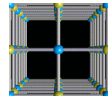
Define body force as follows

$$\mathbf{b}(\mathbf{x}) = \rho \partial_{tt}^2 \mathbf{w}(\mathbf{x}, t) - \mathbf{f}^\epsilon(\mathbf{w}(t))(\mathbf{x})$$

and set initial condition $\mathbf{u}_0(\mathbf{x}) = \mathbf{w}(0, \mathbf{x}), \mathbf{v}_0(\mathbf{x}) = \dot{\mathbf{w}}(0, \mathbf{x})$.

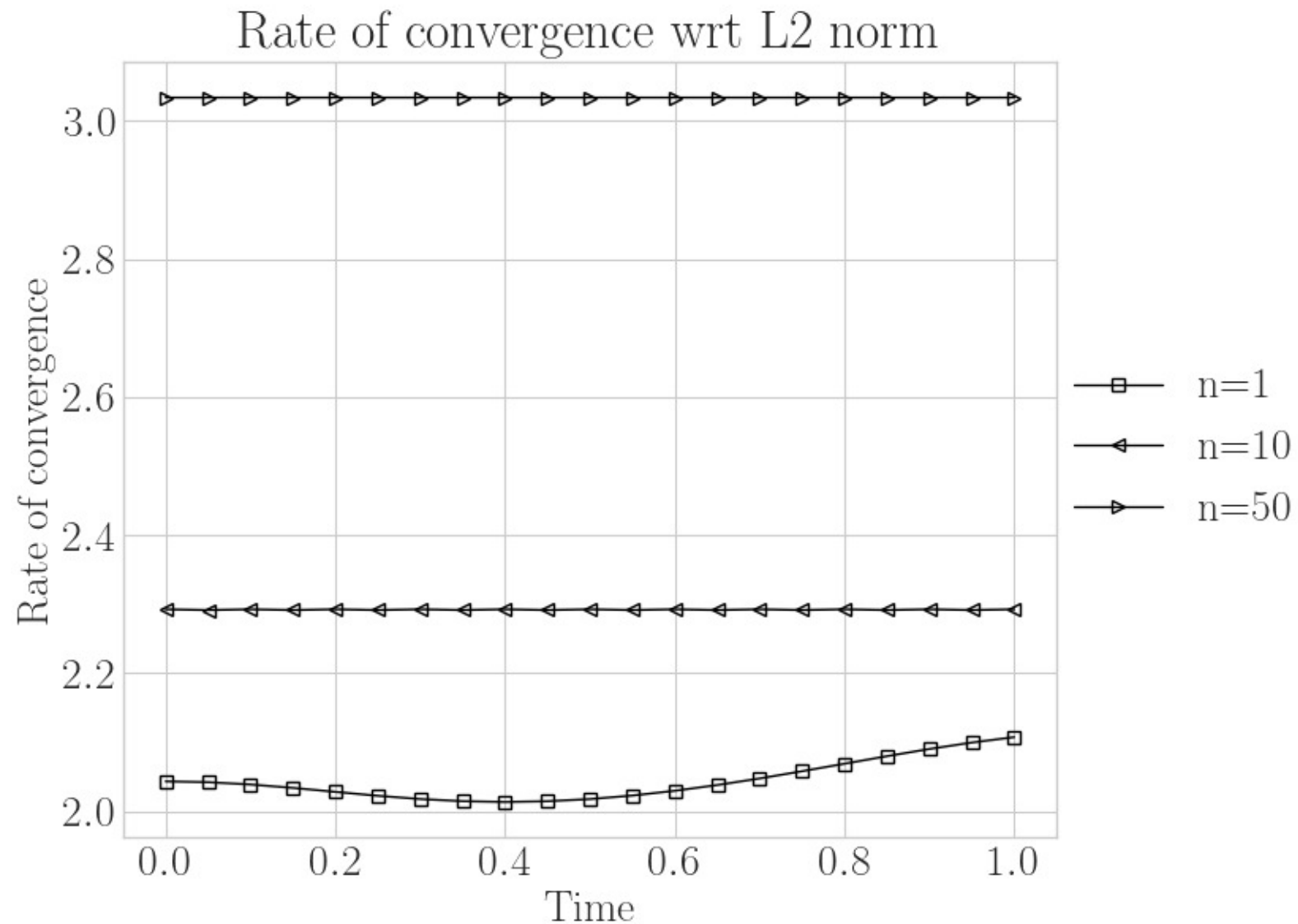
Then $\mathbf{u}(\mathbf{x}, t) = \mathbf{w}(\mathbf{x}, t)$ is the solution.

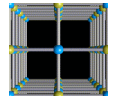




Numerical results: Exact solutions

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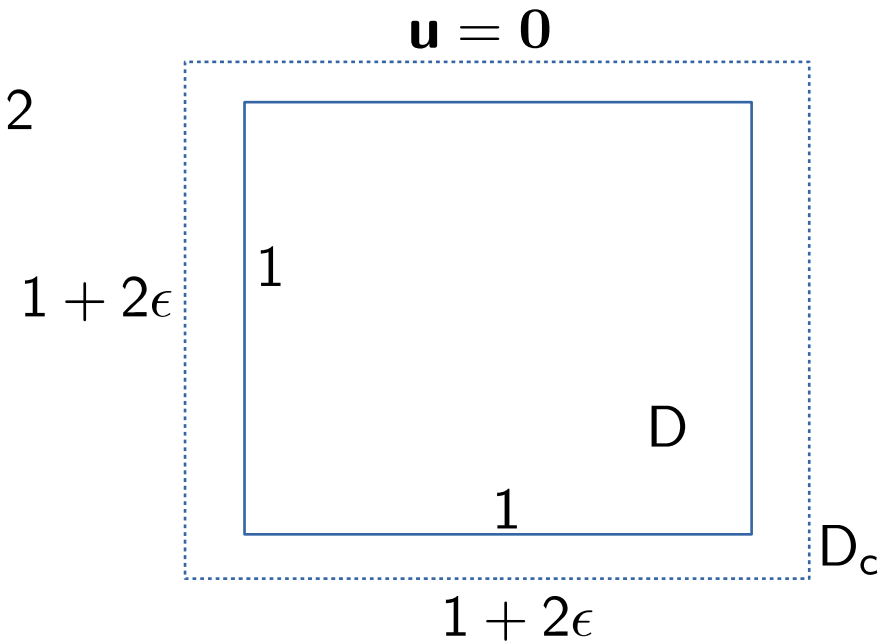


Numerical results: Different initial conditions

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- $\epsilon = 0.2$
- $h = \epsilon/2, \epsilon/4, \epsilon/8$ with $r_h = h_1/h_2 = h_2/h_3 = 2$
- Time domain $[0, 1]$ with $\Delta t = 2 \times 10^{-5}$
- $\rho = 1, f(r) = 1 - \exp[-r], J(r) = 1 - r$

$$\bar{\alpha} := \frac{\log(||\mathbf{u}_1 - \mathbf{u}_2||) - \log(||\mathbf{u}_2 - \mathbf{u}_3||)}{\log(r_h)},$$

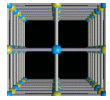


Let $\mathbf{u} = \mathbf{0}$ on D_c . We consider initial condition of the form

$$\mathbf{u}_0(\mathbf{x}) = \mathbf{0}, \quad \mathbf{v}_0(\mathbf{x}) = n\pi a(\mathbf{x}) \exp[-|\mathbf{x} - \mathbf{x}_c|^2/b] \mathbf{d},$$

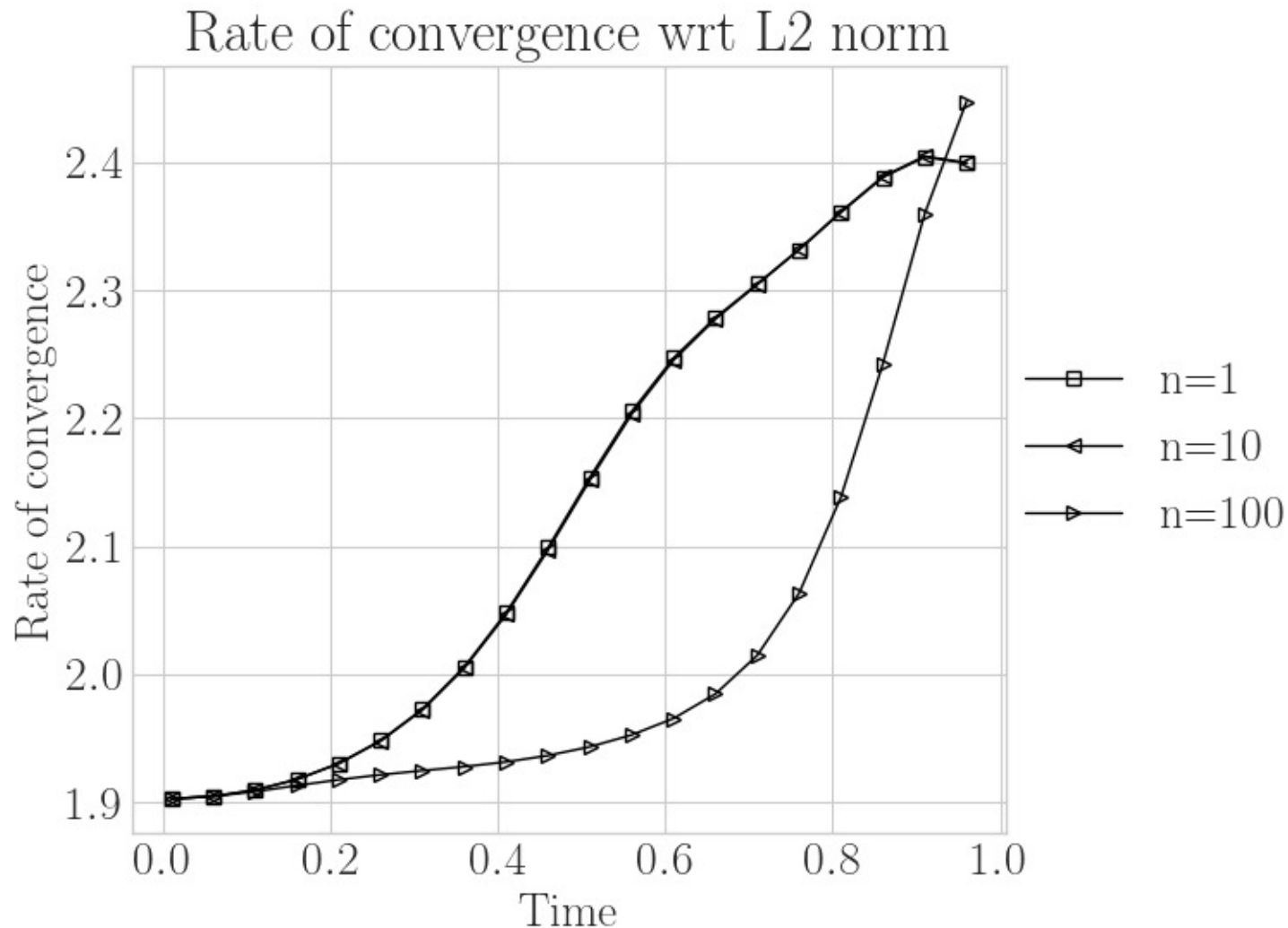
where $a(\mathbf{x}) = 0.1 * x_1 x_2 (1 - x_1)(1 - x_2)$ for $\mathbf{x} \in D$ and 0 otherwise,

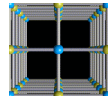
$\mathbf{x}_c = (0.5, 0.5)$, $\mathbf{d} = (0, 1)$, and $b = 0.1$.



Numerical results: Different initial conditions

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Numerical results for damage model

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We introduce new damage model within peridynamic state-based framework in **Lipton et. al. 2018¹**.

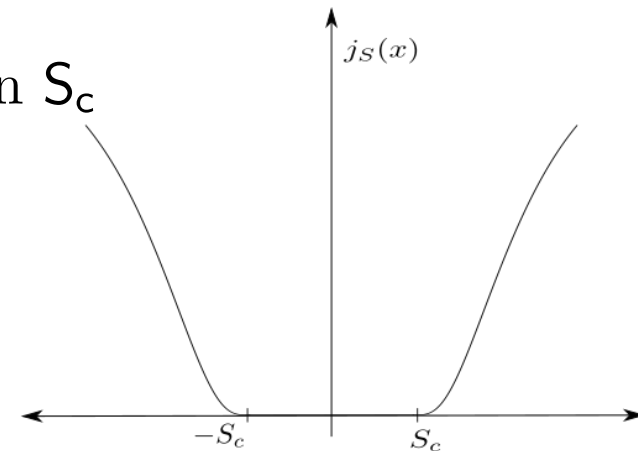
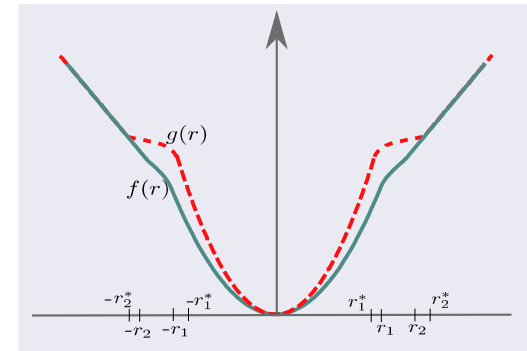
We focus only on bond-based part of the interaction. Peridynamic force is of the form

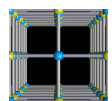
$$\mathbf{f}^\epsilon(\mathbf{x}; \mathbf{u}(t)) = \frac{2}{|B_\epsilon(\mathbf{x})|} \int_{D \cap B_\epsilon(\mathbf{x})} H^T(\mathbf{u})(\mathbf{y}, \mathbf{x}, t) \hat{\mathbf{f}}^\epsilon(\mathbf{y}, \mathbf{x}; \mathbf{u}(t)) d\mathbf{y},$$

where damage of bond $\mathbf{y} - \mathbf{x}$ at time t is given by

$$H^T(\mathbf{u})(\mathbf{y}, \mathbf{x}, t) = h \left(\underbrace{\int_0^t j_S(S(\mathbf{y}, \mathbf{x}, \tau; \mathbf{u})) d\tau}_{\text{damage integral}} \right)$$

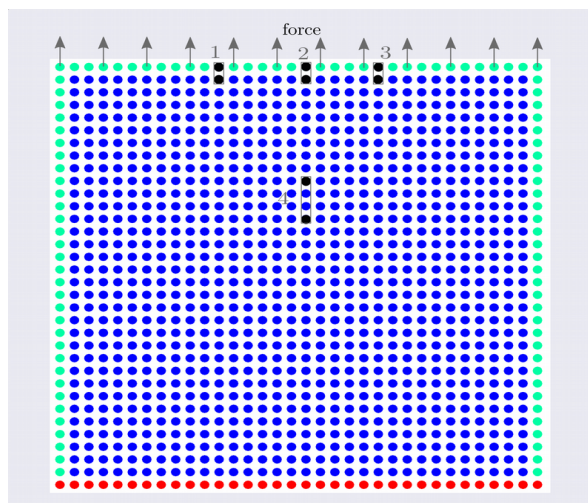
j_S is nonzero positive for strain only above critical strain S_c



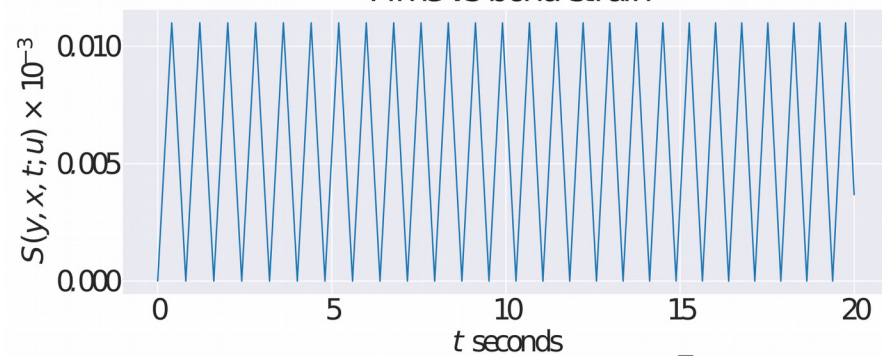


Periodic loading

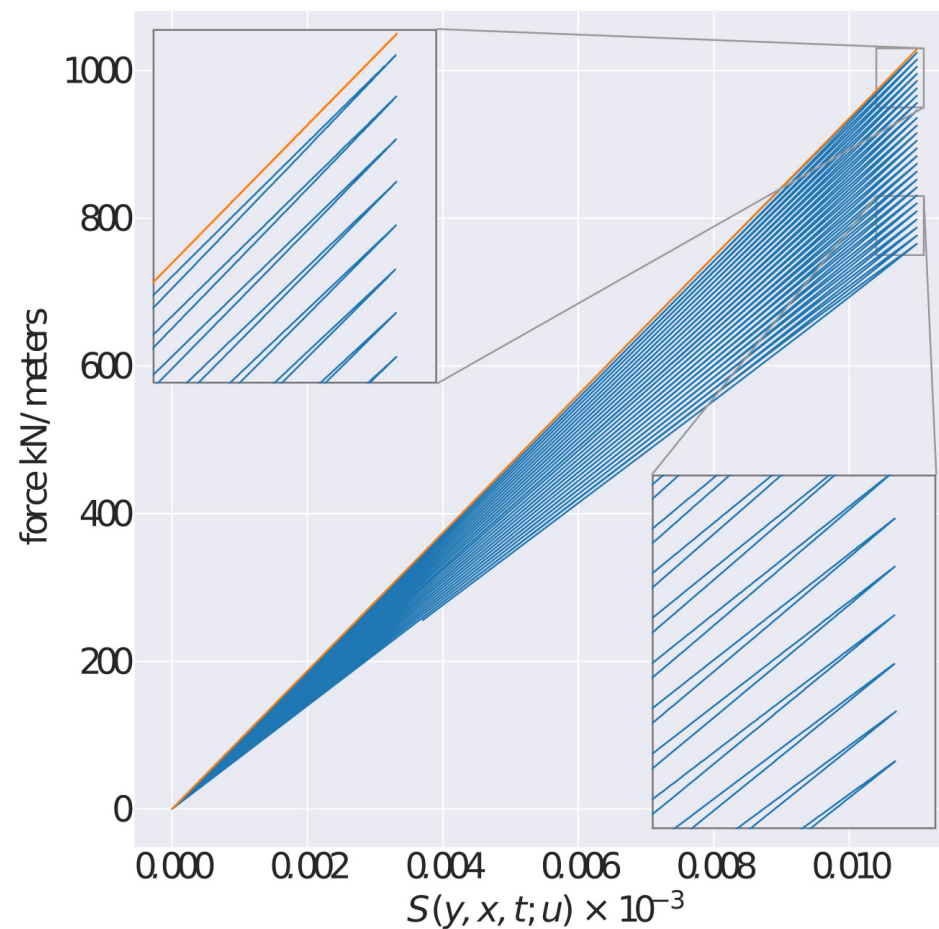
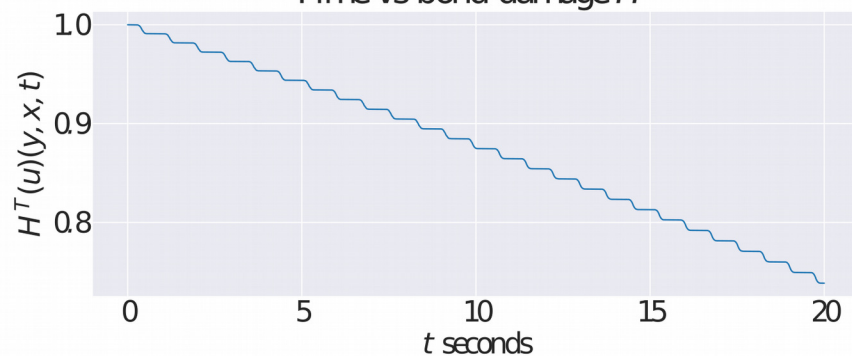
32

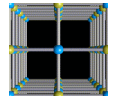


Time vs bond-strain



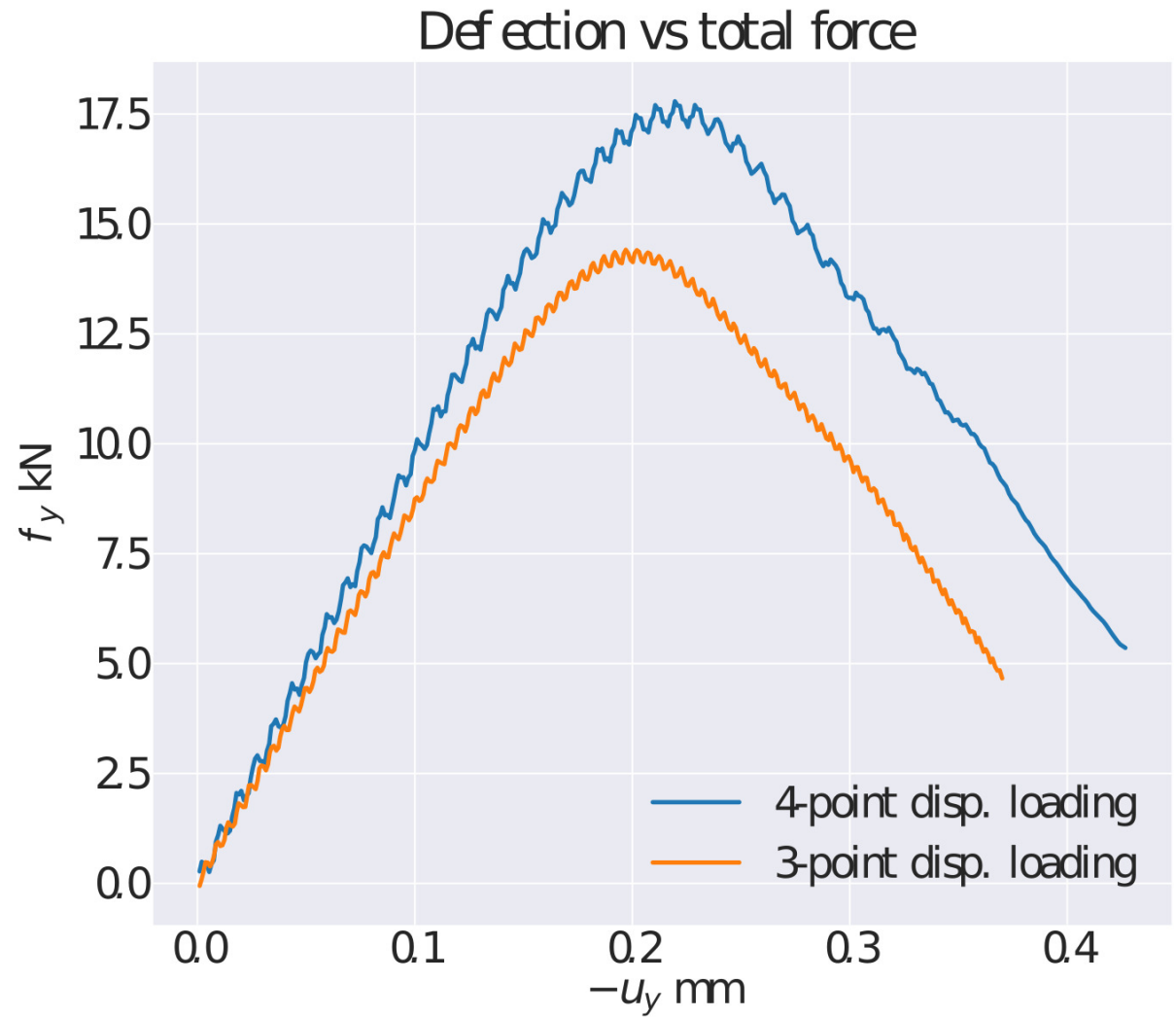
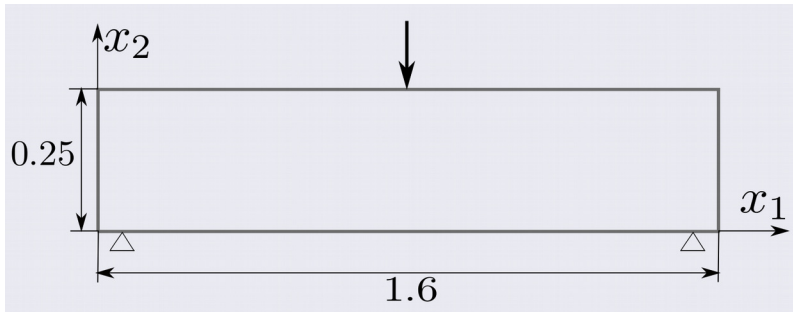
Time vs bond-damage H^T

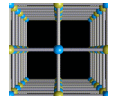




Bending test

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Future works

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- Numerical analysis of state-based model.
- Analysis of state-based peridynamic energy in the limit nonlocal length-scale tends to zero.
- Implementation of adaptive-mesh refinement.

Thank you!